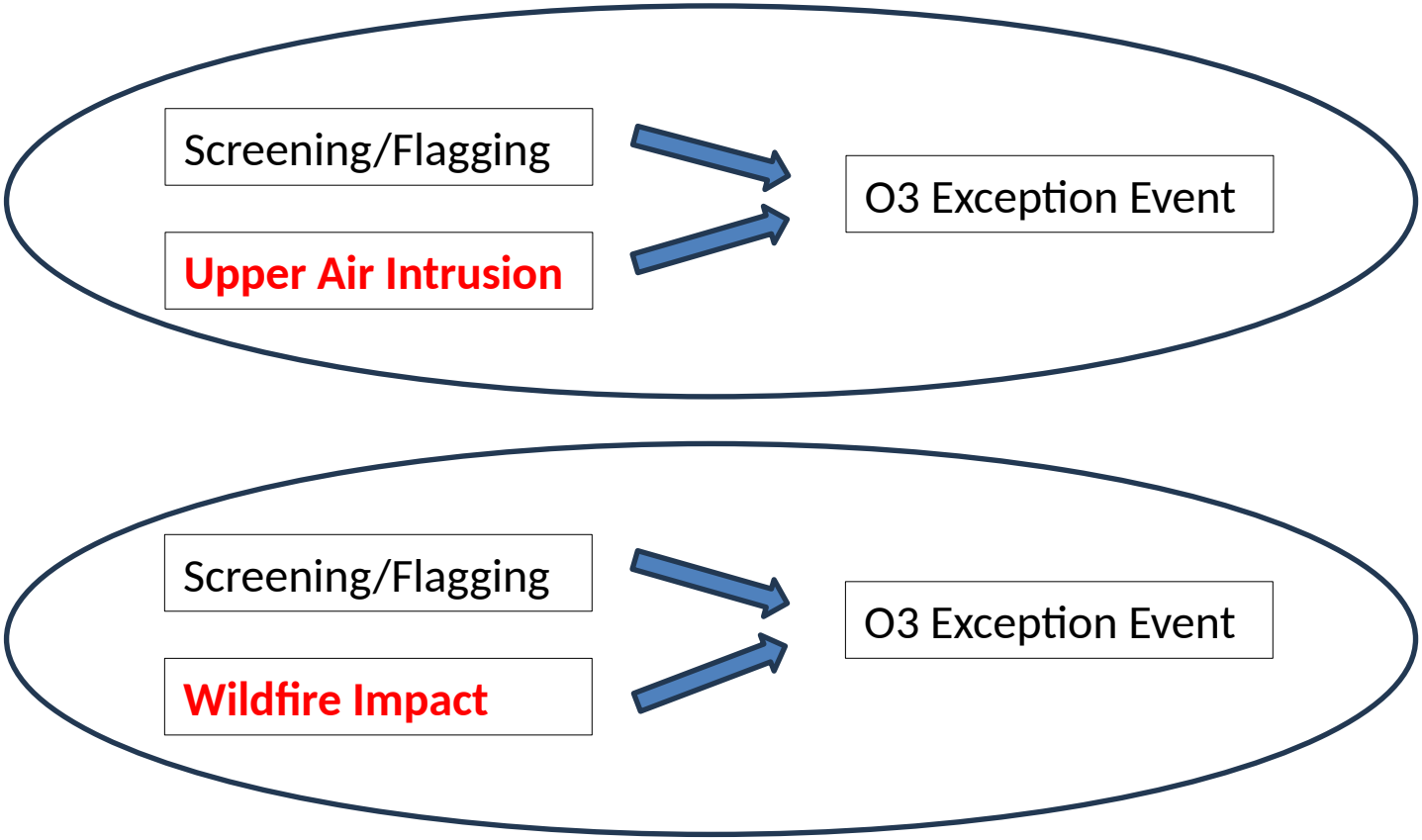
Using a Machine Learning Model for Ozone Exceptional Events Screening

02/06/2026

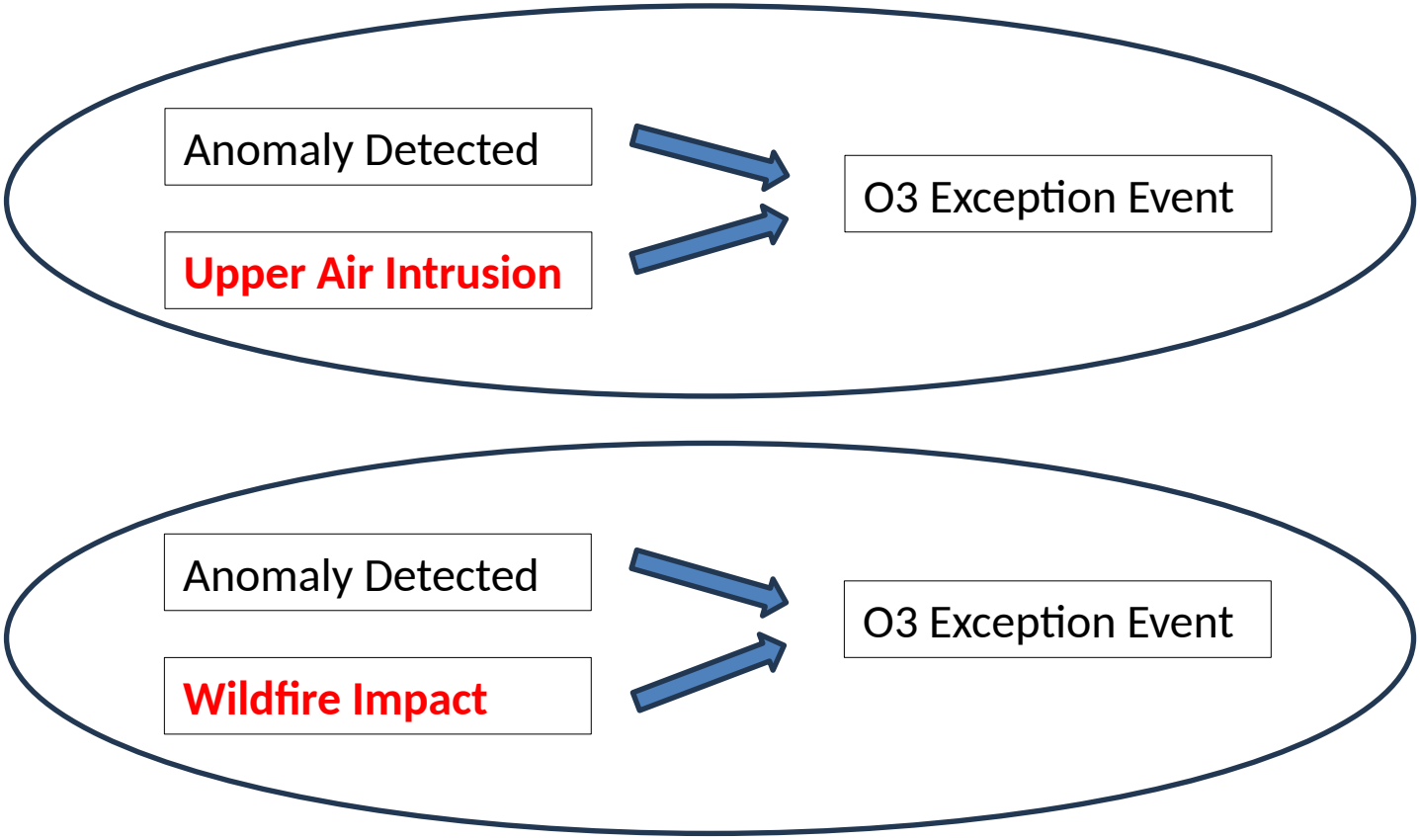
O3 Exception Events

- Unusual or naturally occurring events that can affect air quality but are not reasonably controllable using techniques that tribal, state or local air agencies may implement in order to attain and maintain the National Ambient Air Quality Standards.
- Common Sources of O3 Exception Events
  - Upper Air Intrusion
  - Wildfire Impact

Identify O3 Exception Events



Using Machine Learning Model as Screening Tool



Machine Learning Model

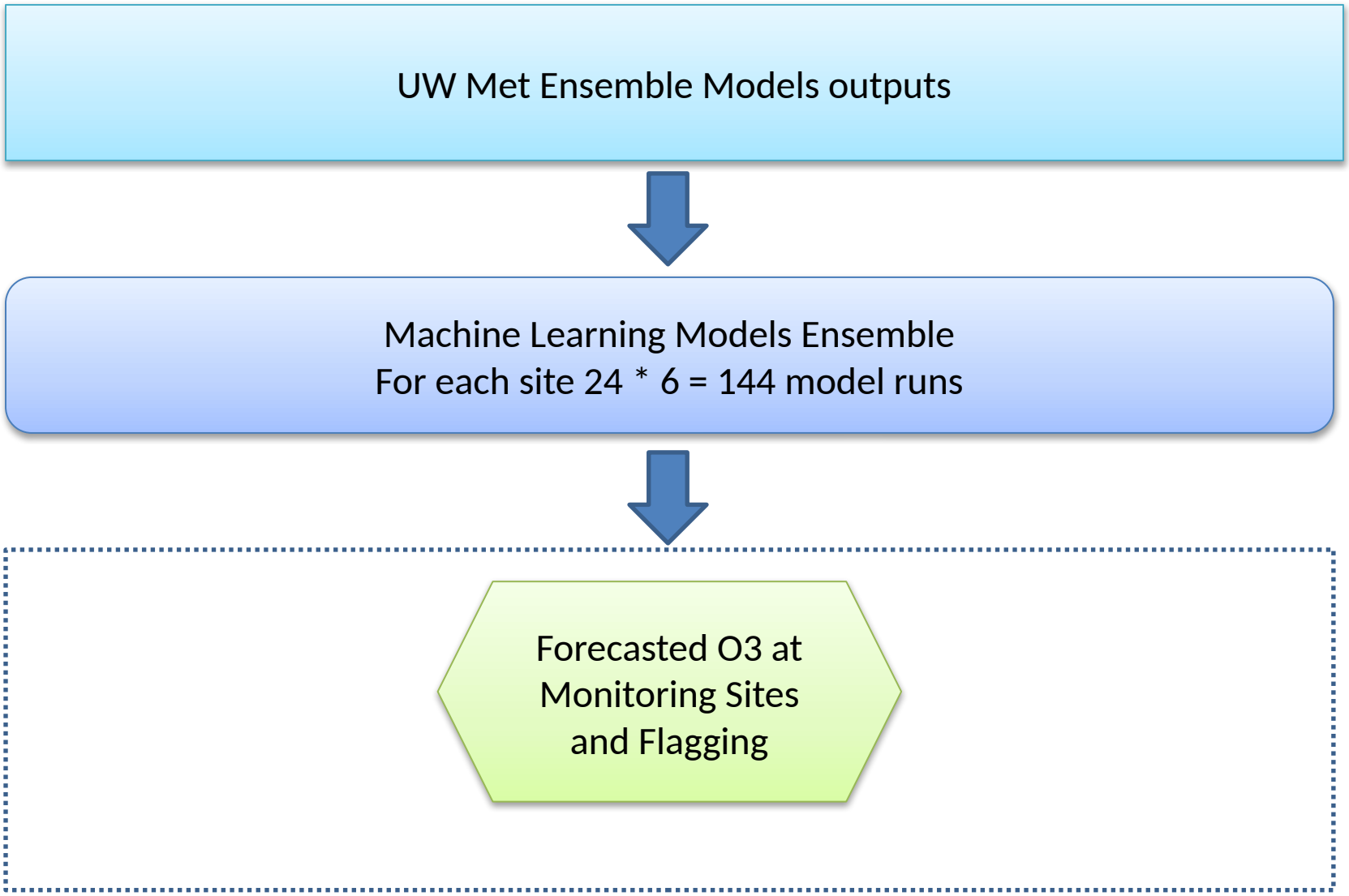
- Build a machine learning model that is only influenced by normal emission and meteorology.

- Remove wildfire days from training and testing data set
  - Eliminate the impact from wildfire events

Key Assumption

- Emissions are roughly constant with monthly, weekday/weekend, and hourly variation.
- Anomalies, such as upper air intrusion events, are infrequently for each specific site crossing multiple years.
- Wildfire events are removed from training and test data set

Overview of Forecast System



O3 Forecast
Machine Learning Models Used

Neural Network Models

- Dense Neural Network Model
- 1D Convolutional Neural Network Model
- Recurrent Neural Network (LSTM)

Decision Tree Based Models

- Pure XGBoost
- Random Forest
- Boosted Random Forest

O3 Forecast
Input Parameters Used

10 Input Parameters :

- T, P, RH, PBLH, WS, WDIR
- O3hourly_mean (grouped by Month, Weekday/end, Hour) excluding wildfire days
- Month (1-12), Weekday (0-1), Hour (0-23)

Training for 2024 O3 Forecast

- Training data set:
  - 2017 - 2021
- Testing data set:
  - 2022 – 2023

- Removed wildfire days from training and testing data set

Performance

TEST							
St. Lukes Meridian (AQSID 160010010)							
Forecast	**max_error**	**mean_absolute_error**	**mean_squared_error**	**normalized_mean_bias**	**normalized_mean_error**	**r2_score**	**root_mean_squared_error**
ensemble 05	23	4.58	32.74	-0.07	0.11	0.76	5.72
White Pine (AQSID 160010017)							
Forecast	**max_error**	**mean_absolute_error**	**mean_squared_error**	**normalized_mean_bias**	**normalized_mean_error**	**r2_score**	**root_mean_squared_error**
ensemble 05	24	6.1	56.35	-0.08	0.13	0.26	7.51
Pocatello Garrett and Gould (AQSID 160050015)							
Forecast	**max_error**	**mean_absolute_error**	**mean_squared_error**	**normalized_mean_bias**	**normalized_mean_error**	**r2_score**	**root_mean_squared_error**
ensemble 05	38	4.47	35.89	-0.03	0.1	0.31	5.99

Flags for O3

- Only for days:
  - Observation of O3 8hr Avg Max > 54 ppb
- Flag 1
  - -10 <= Bias (Prediction - Observation) < -5
- Flag 2
  - -15 <= Bias (Prediction - Observation) < -10
- Flag 3
  - -20 <= Bias (Prediction - Observation) < -15
- Flag 4
  - Bias (Prediction - Observation) < -20

Running for 2024

- Total sites: 3 O3 sites
  - 160010010 : Meridian
  - 160010017 : White Pine
  - 160050015 : Pocatello
- AM and PM forecast for every day

Evaluating for 2024

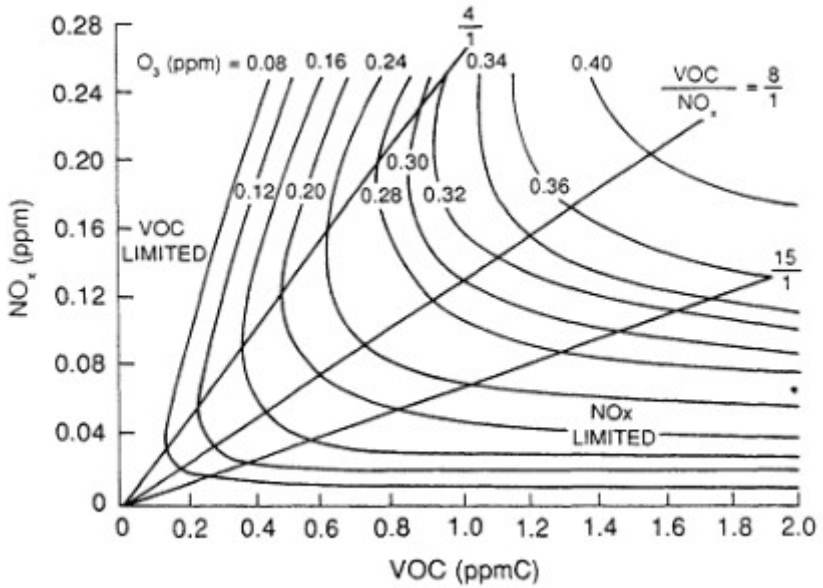
- Using PM day 1 result
- Flags of days
- Distribution of O3 for each Flag
- Observation vs. PM day 1 time series
  - Hourly
  - Daily
- Daily Bias time series

Wildfire Impact on O3

- The effect of wildfire impact on O3 vary
  - VOC, NOX
  - Block Sunlight
  - Changing VOC/NOX ratio

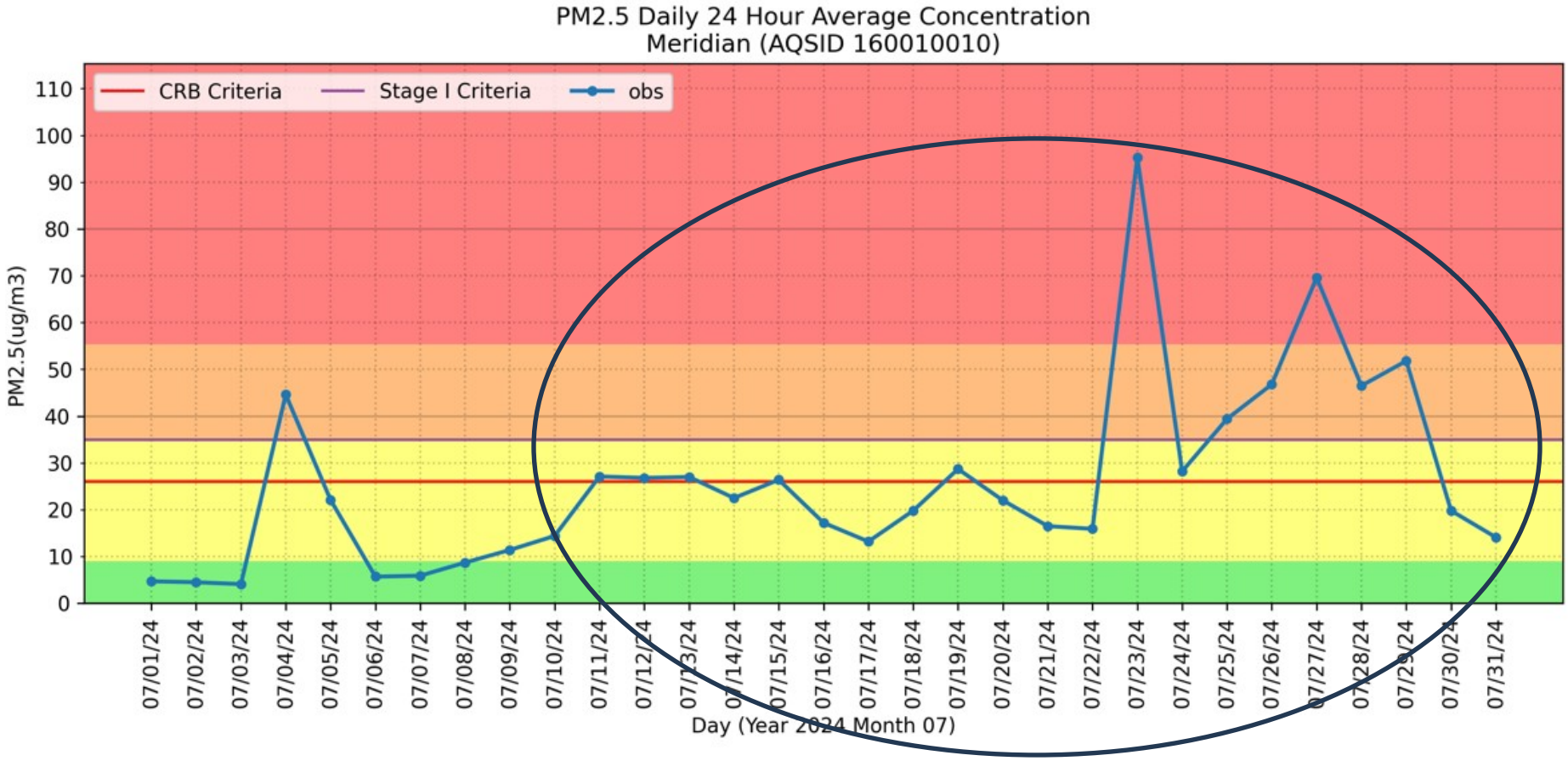
- Wildfire can increase O3 production
- Wildfire can decrease O3 production

VOC and NOx: Relationship to Ozone

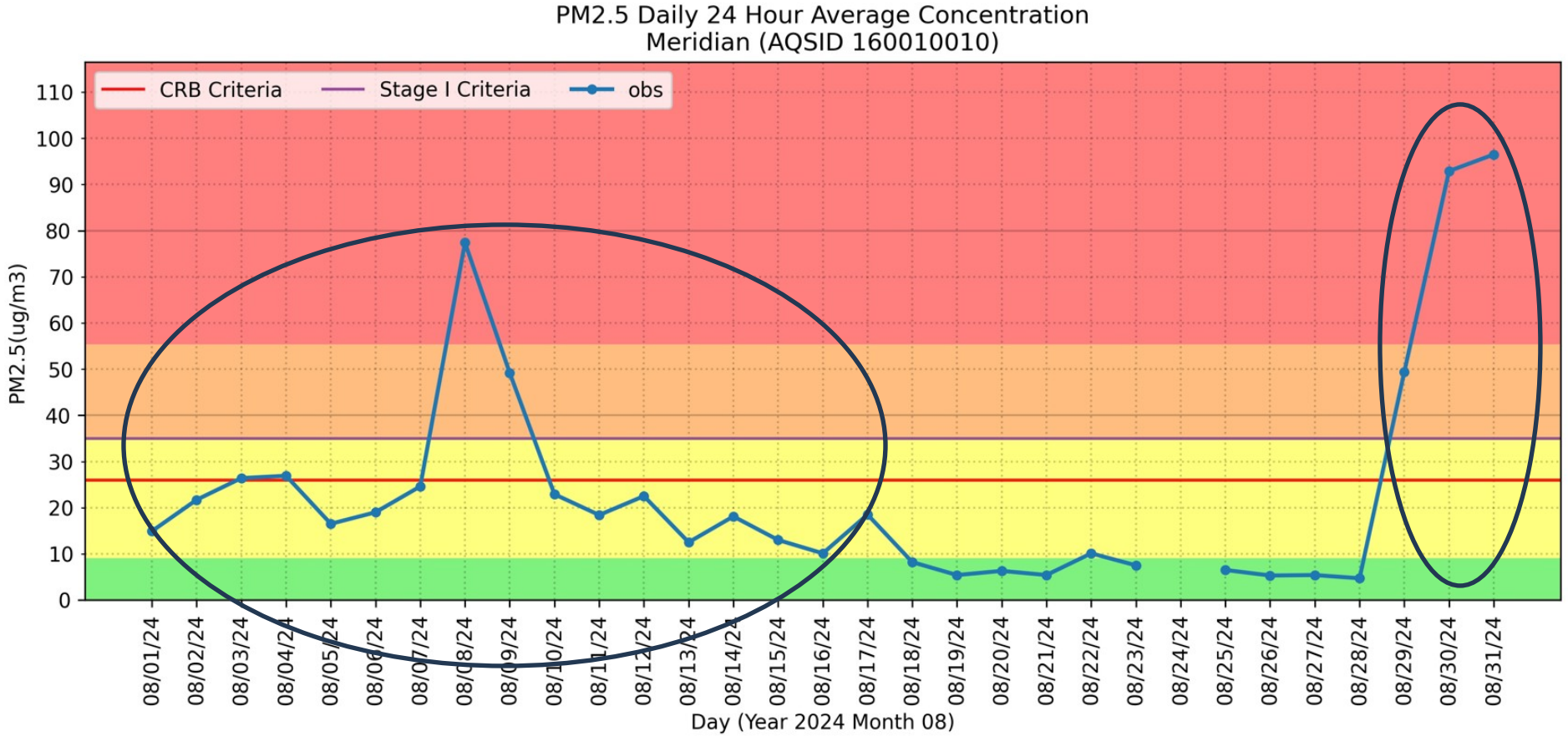


National Academies of Sciences, Engineering, and Medicine. 1991. Rethinking the Ozone Problem in Urban and Regional Air Pollution. Washington, DC: The National Academies Press. https://doi.org/10.17226/1889.

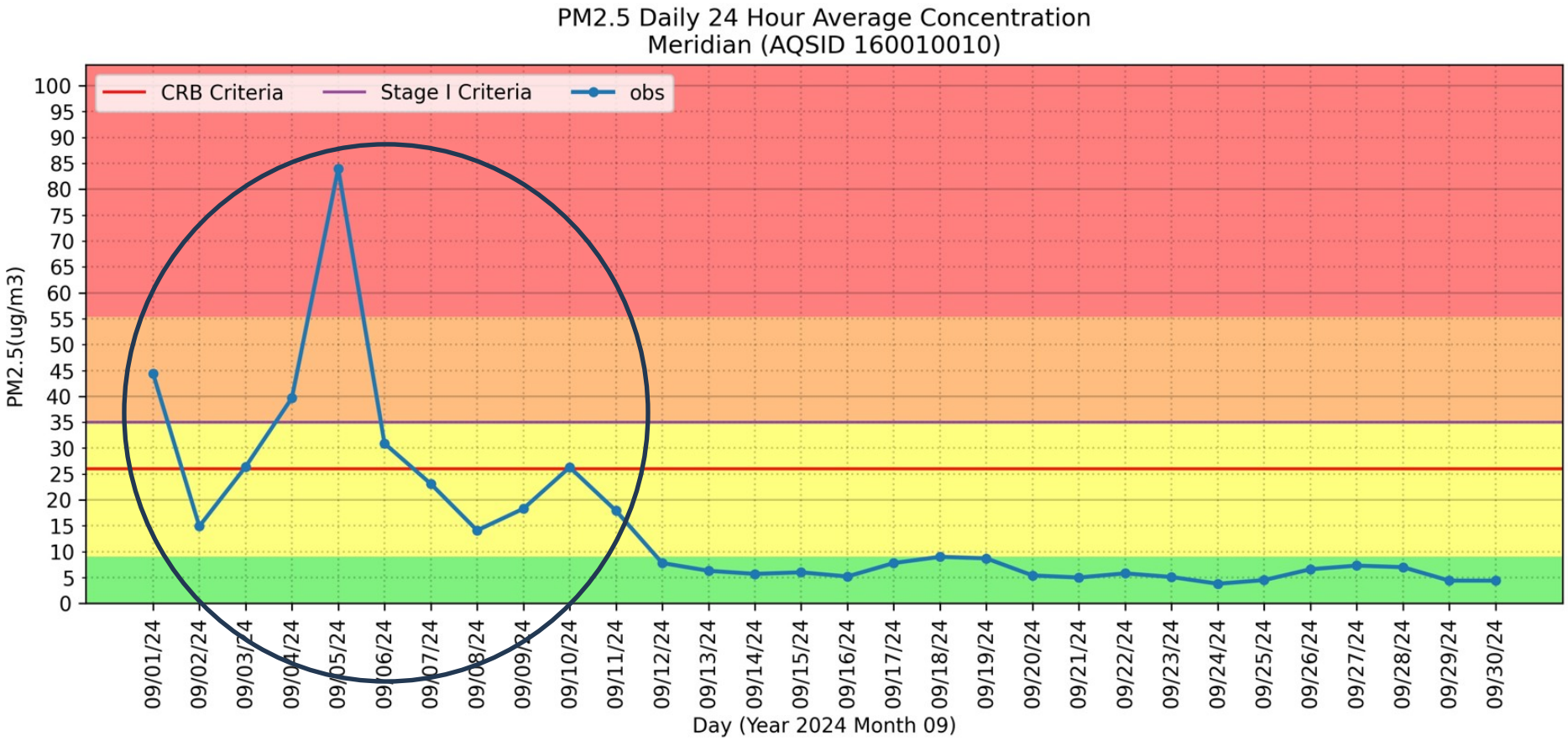
July Wildfire Impact at Meridian Site



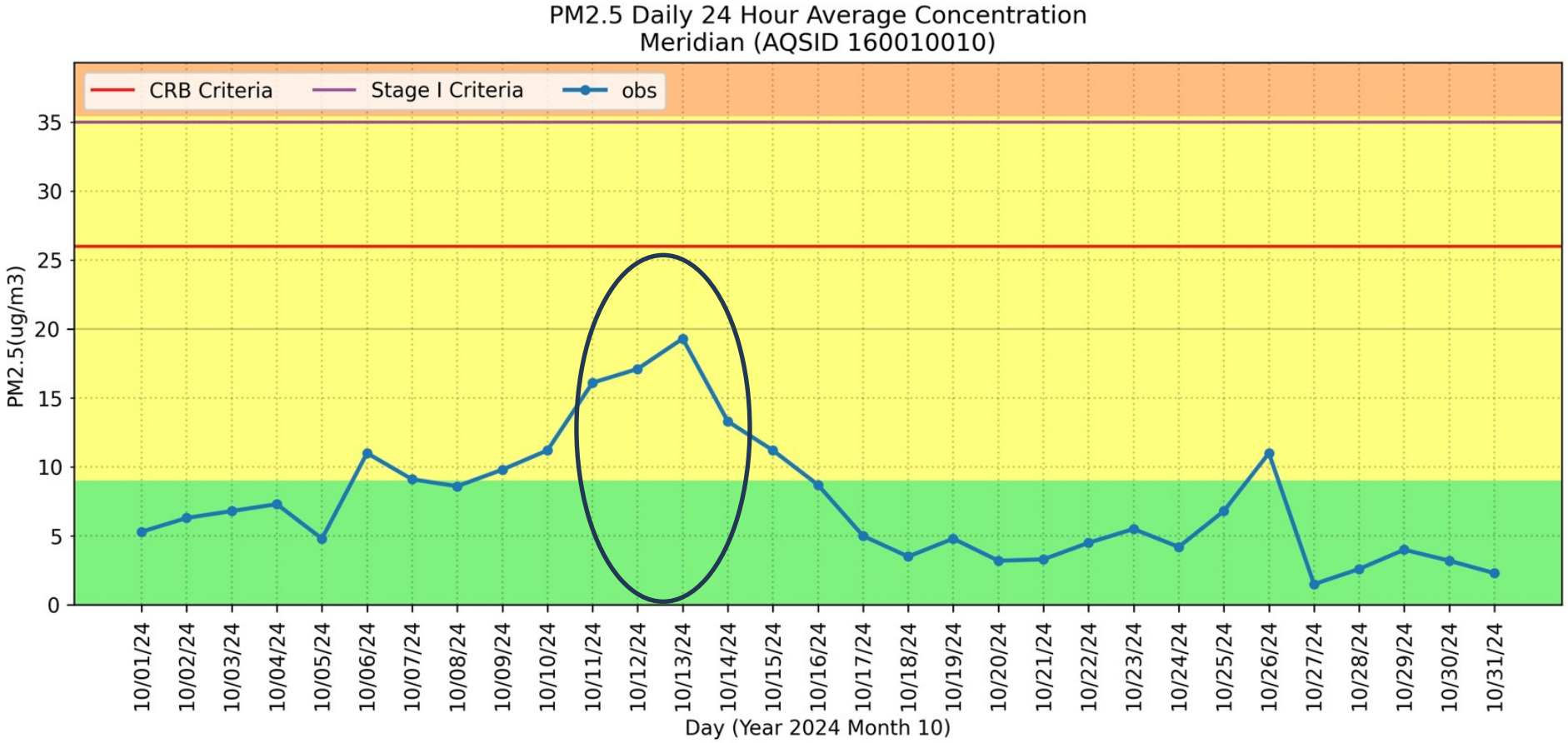
August Wildfire Impact at Meridian Site



September Wildfire Impact at Meridian Site



October Wildfire Impact at Meridian Site



Flags of Meridian Site

Flag 1

LocalDateTime	obs	pure_model_pm_day1	bias
2024-04-21	55	47	-8
2024-04-22	56	47	-9
2024-04-23	56	47	-9
2024-05-10	55	48	-7
2024-05-11	57	49	-8
2024-05-15	56	49	-7
2024-05-18	56	50	-6
2024-05-27	57	50	-7
2024-05-28	56	49	-7
2024-05-30	55	48	-7
2024-05-31	56	49	-7
2024-06-06	57	50	-7
2024-06-22	66	56	-10
2024-06-25	62	55	-7
2024-07-03	58	49	-9
2024-07-04	61	53	-8
2024-07-09	68	60	-8
2024-07-10	69	59	-10
2024-07-14	65	58	-7
2024-07-15	63	56	-7
2024-07-17	65	57	-8
2024-07-21	70	60	-10
2024-07-27	61	54	-7
2024-07-29	59	52	-7
2024-08-03	69	59	-10
2024-08-06	63	57	-6
2024-08-10	62	54	-8
2024-08-27	63	53	-10
2024-09-02	58	51	-7
2024-09-09	56	50	-6

Flag 2

LocalDateTime	obs	pure_model_pm_day1	bias
2024-05-12	59	48	-11
2024-06-05	62	47	-15
2024-06-11	65	52	-13
2024-06-12	65	52	-13
2024-06-14	62	50	-12
2024-07-08	73	58	-15
2024-07-11	71	58	-13
2024-07-16	70	58	-12
2024-07-18	73	59	-14
2024-07-19	72	59	-13
2024-07-20	74	59	-15
2024-07-22	73	59	-14
2024-07-24	73	60	-13
2024-07-28	66	51	-15
2024-08-01	73	59	-14

Flag 3

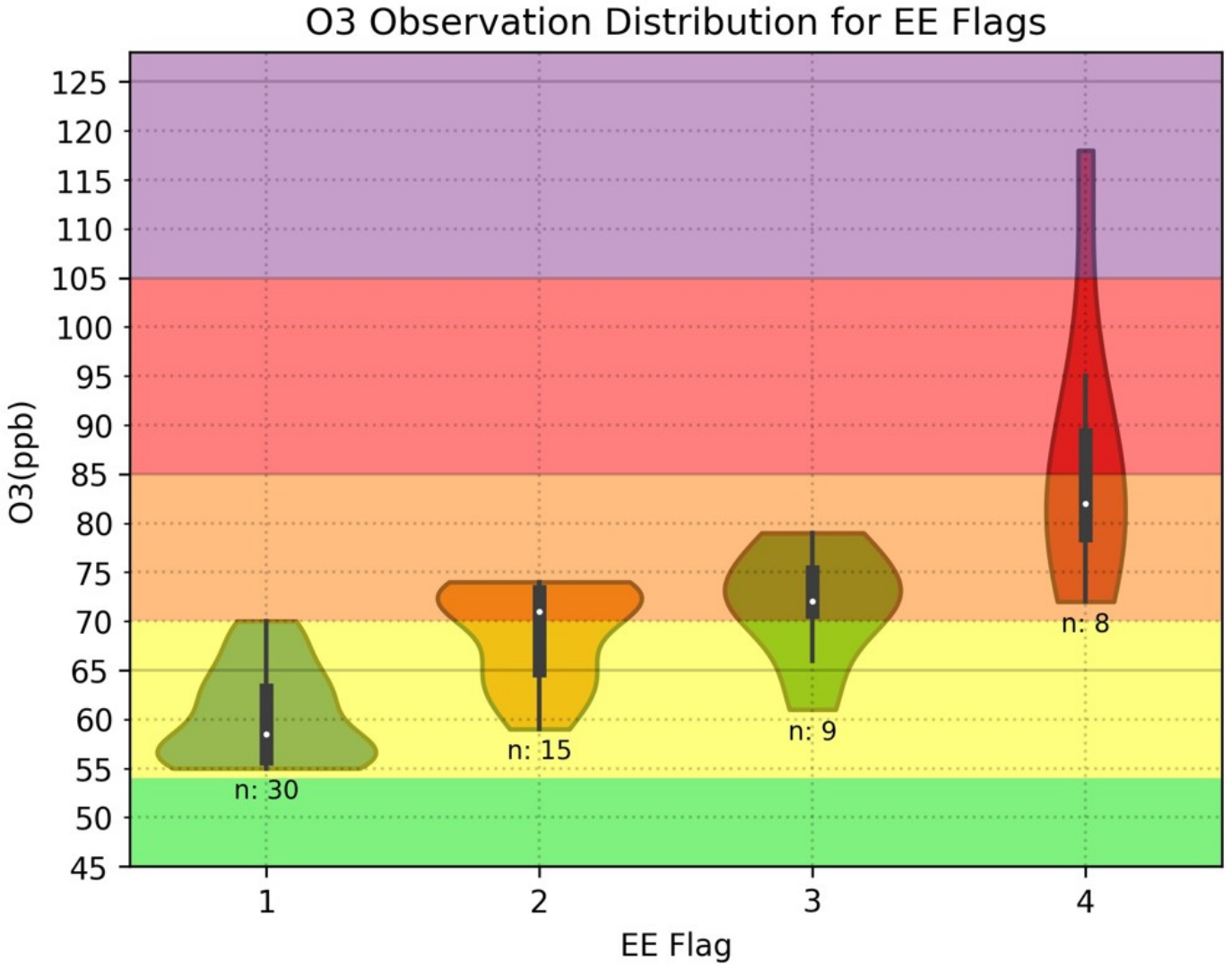
LocalDateTime	obs	pure_model_pm_day1	bias
2024-04-14	61	43	-18
2024-07-12	79	59	-20
2024-07-26	71	53	-18
2024-08-02	78	59	-19
2024-08-07	75	55	-20
2024-08-09	72	56	-16
2024-08-30	73	56	-17
2024-09-01	72	56	-16
2024-09-04	66	47	-19

Flag 4

LocalDateTime	obs	pure_model_pm_day1	bias
2024-06-13	79	55	-24
2024-07-13	83	60	-23
2024-07-23	118	57	-61
2024-07-25	81	52	-29
2024-08-08	95	56	-39
2024-08-31	87	56	-31
2024-09-05	72	50	-22
2024-09-06	78	52	-26

Wildfire impact days are highlighted in yellow

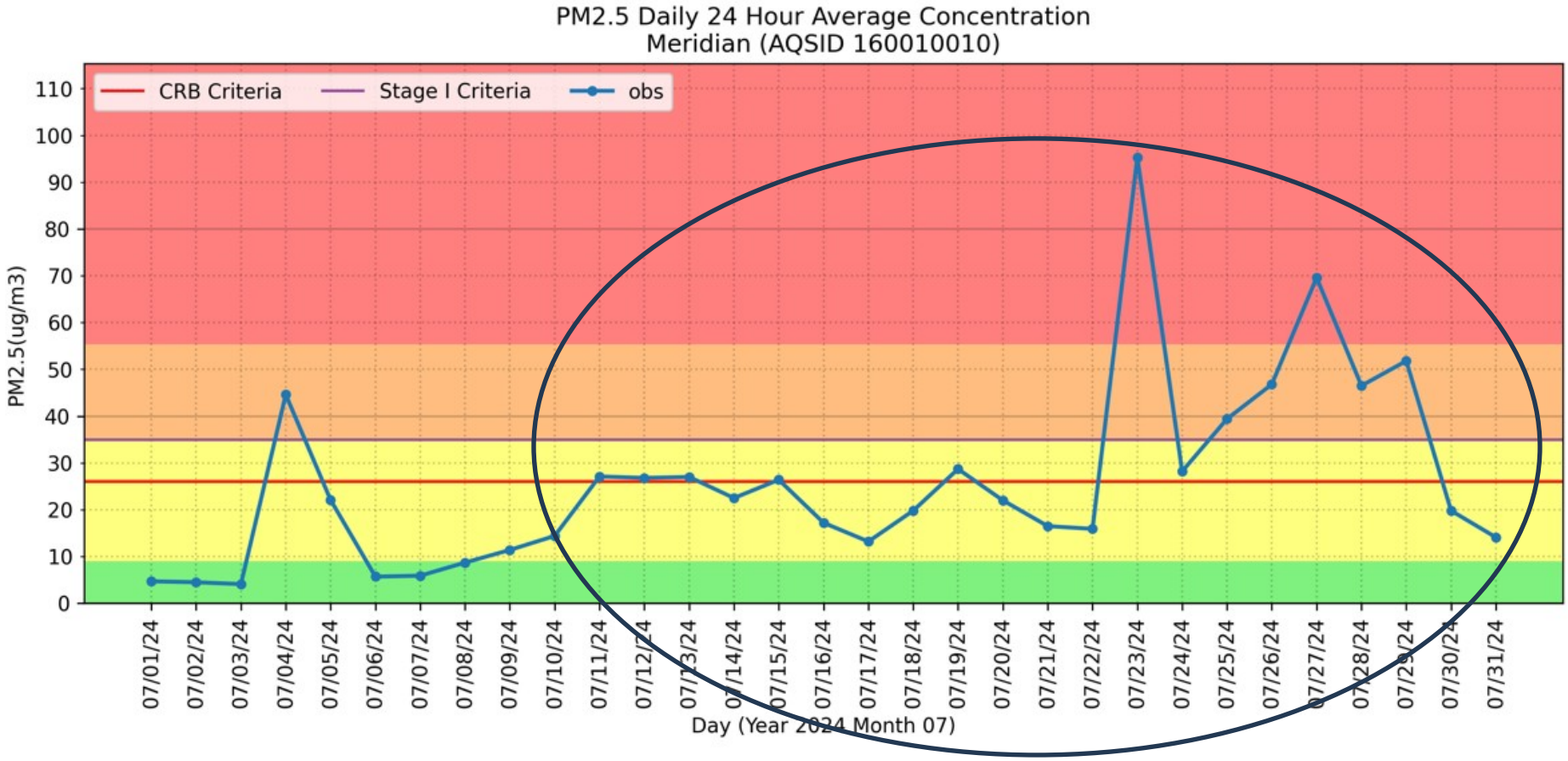
O3 Distribution for Flags at Meridian Site



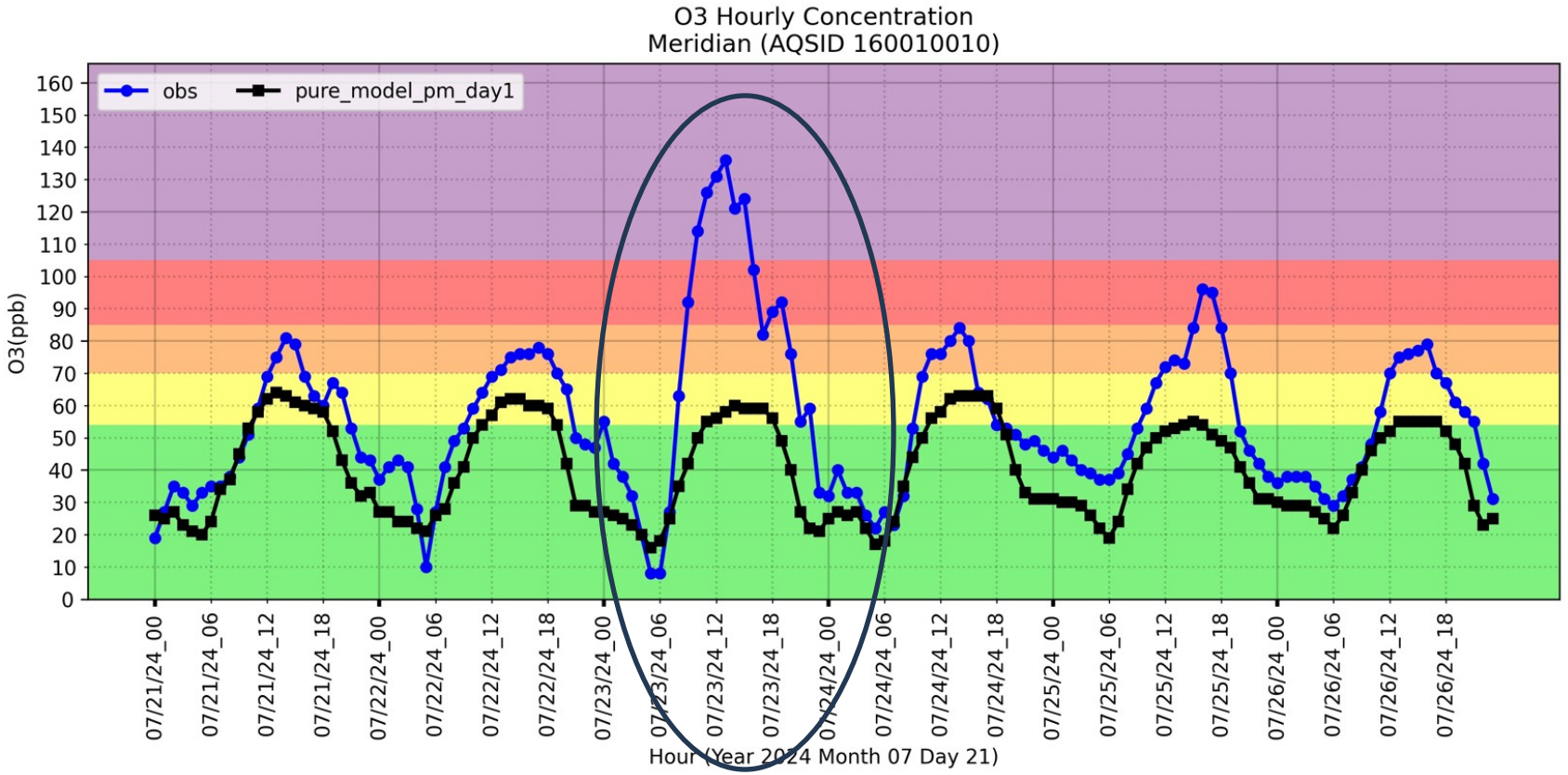
Unflagged Wildfire Impact Days at Meridian Site

Unflagged Wildfire Impact days			
LocalDateTime	**obs**	**pure_model_pm_day1**	**bias**
2024-07-30	41	41	0
2024-07-31	56	54	-2
2024-08-04	58	53	-5
2024-08-05	55	56	1
2024-08-11	59	55	-4
2024-08-12	52	56	4
2024-08-13		51	
2024-08-14	53	51	-2
2024-08-15		52	
2024-08-17	55	55	0
2024-08-29	56	53	-3
2024-09-03	53	45	-8
2024-09-07	59	55	-4
2024-09-08	53	55	2
2024-09-10	55	51	-4
2024-09-11	36	39	3
2024-10-11	39	38	-1
2024-10-12	44	41	-3
2024-10-13	40	39	-1
2024-10-14	43	39	-4

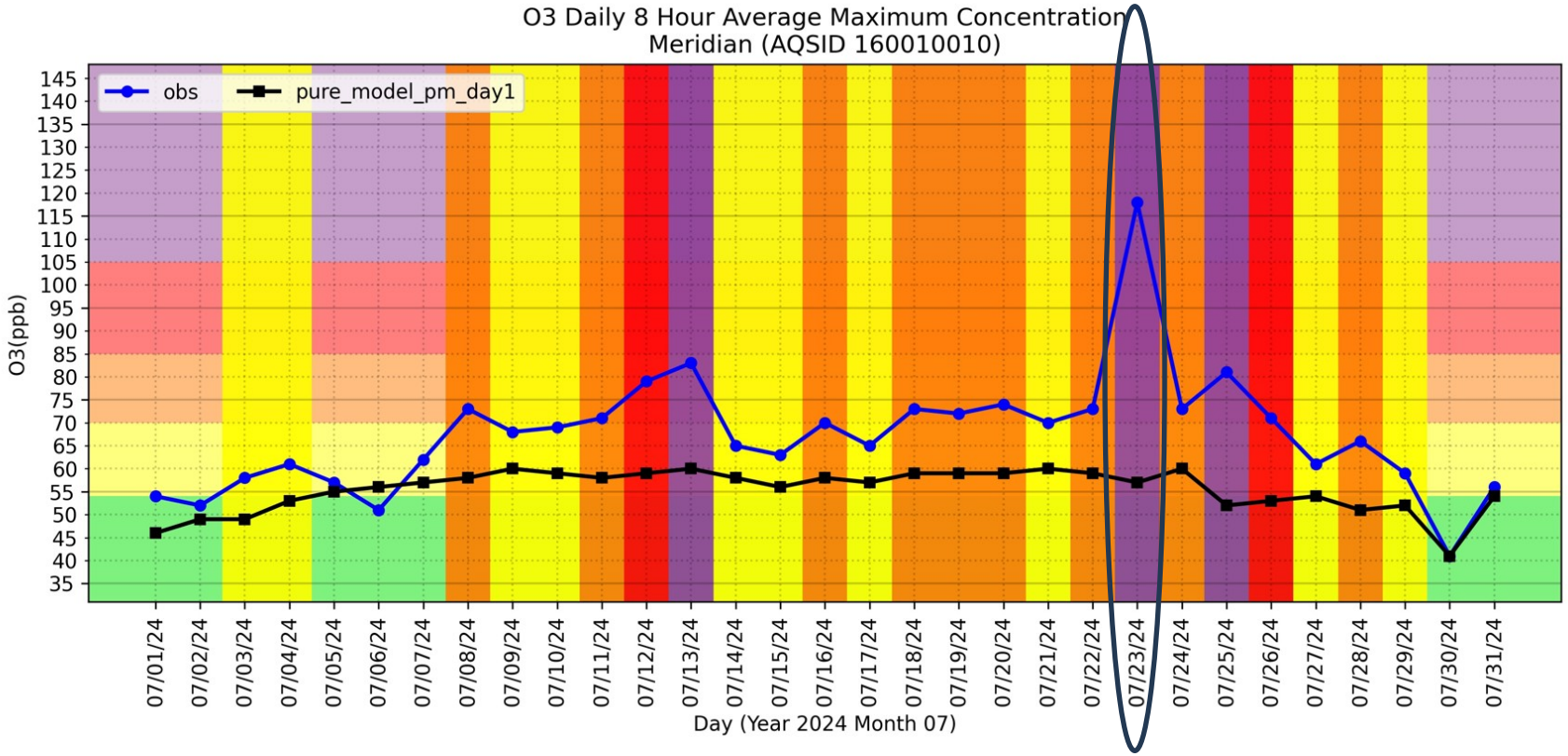
July 23 at Meridian Site (0)
Wildfire Impact



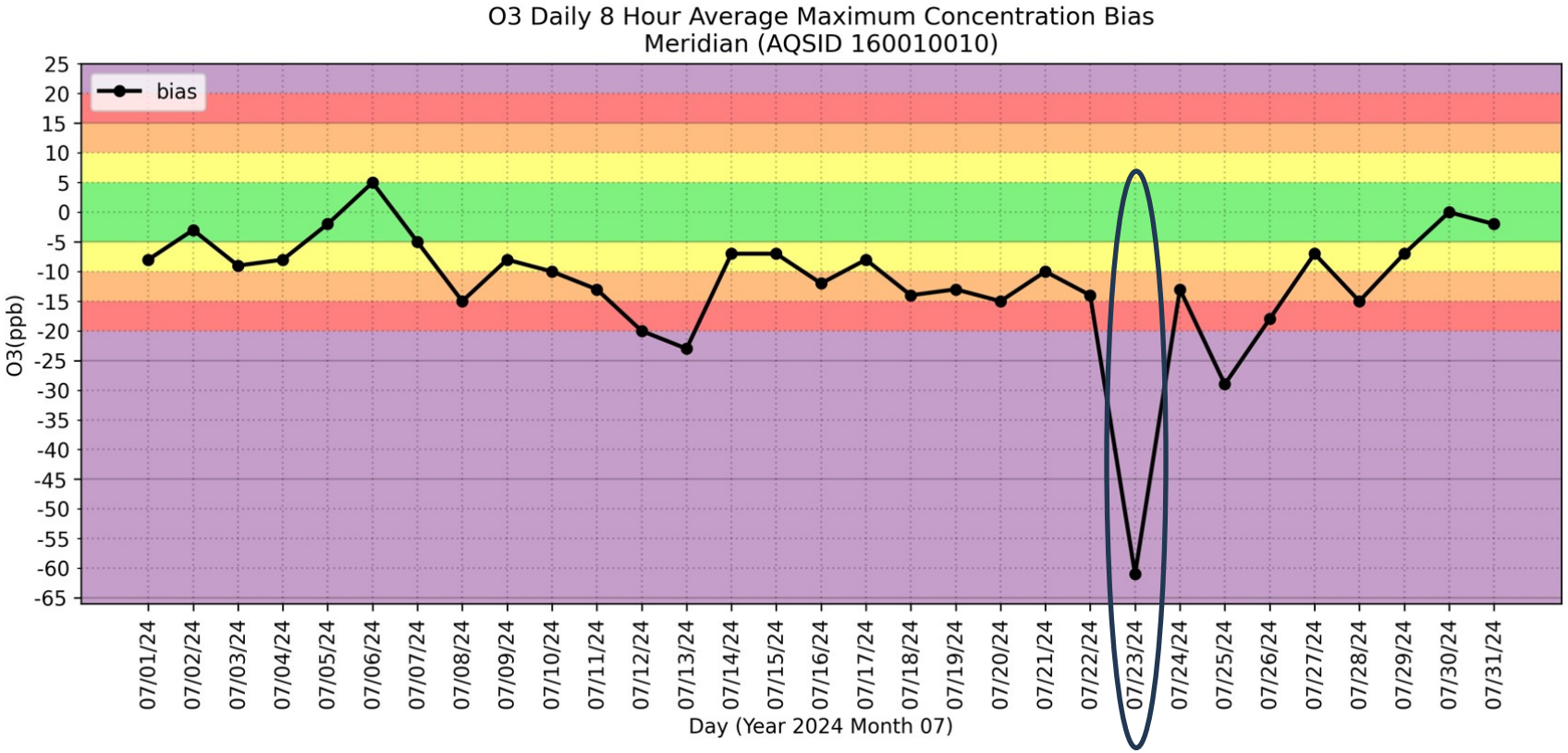
July 23 at Meridian Site (1) Wildfire Impact



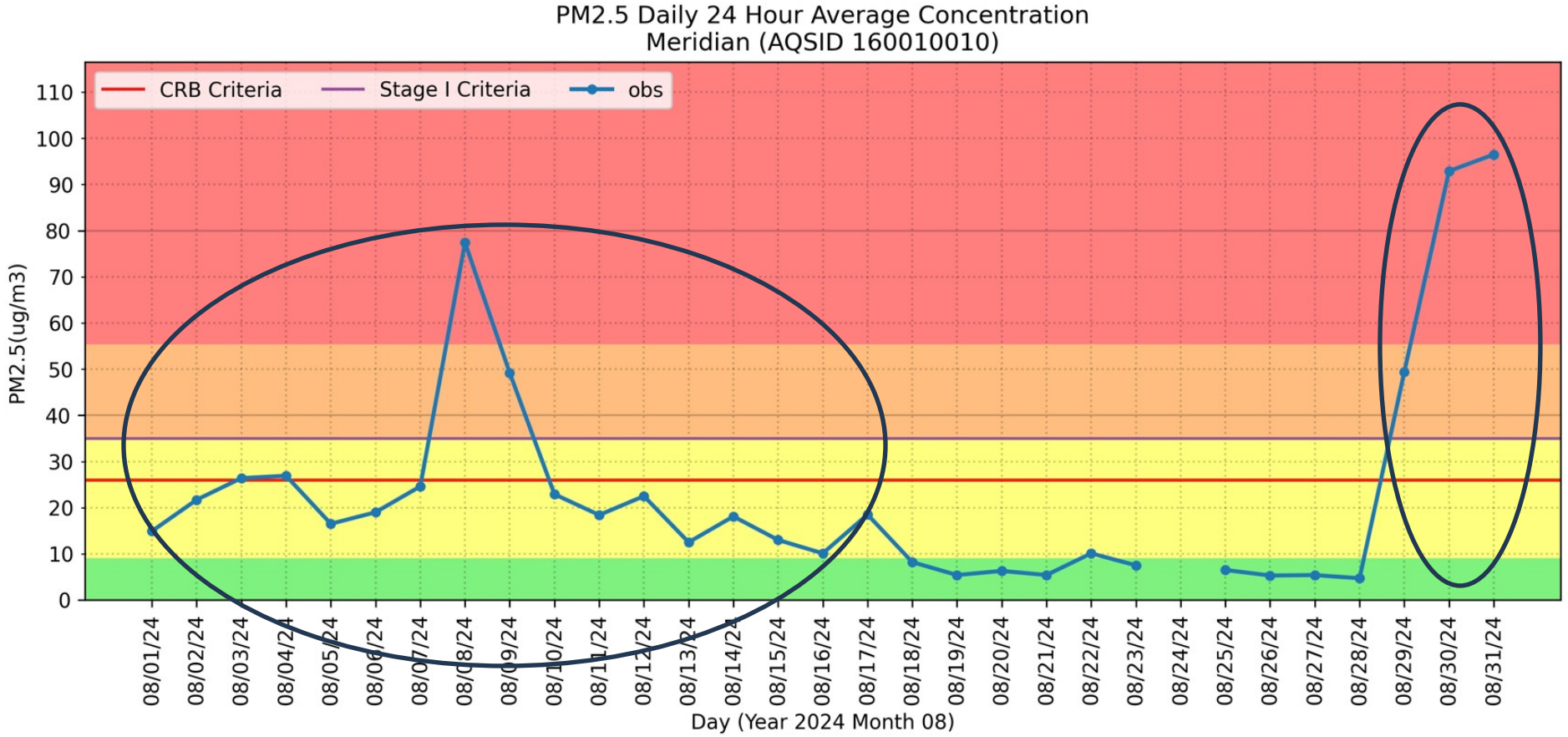
July 23 at Meridian Site (2) Wildfire Impact



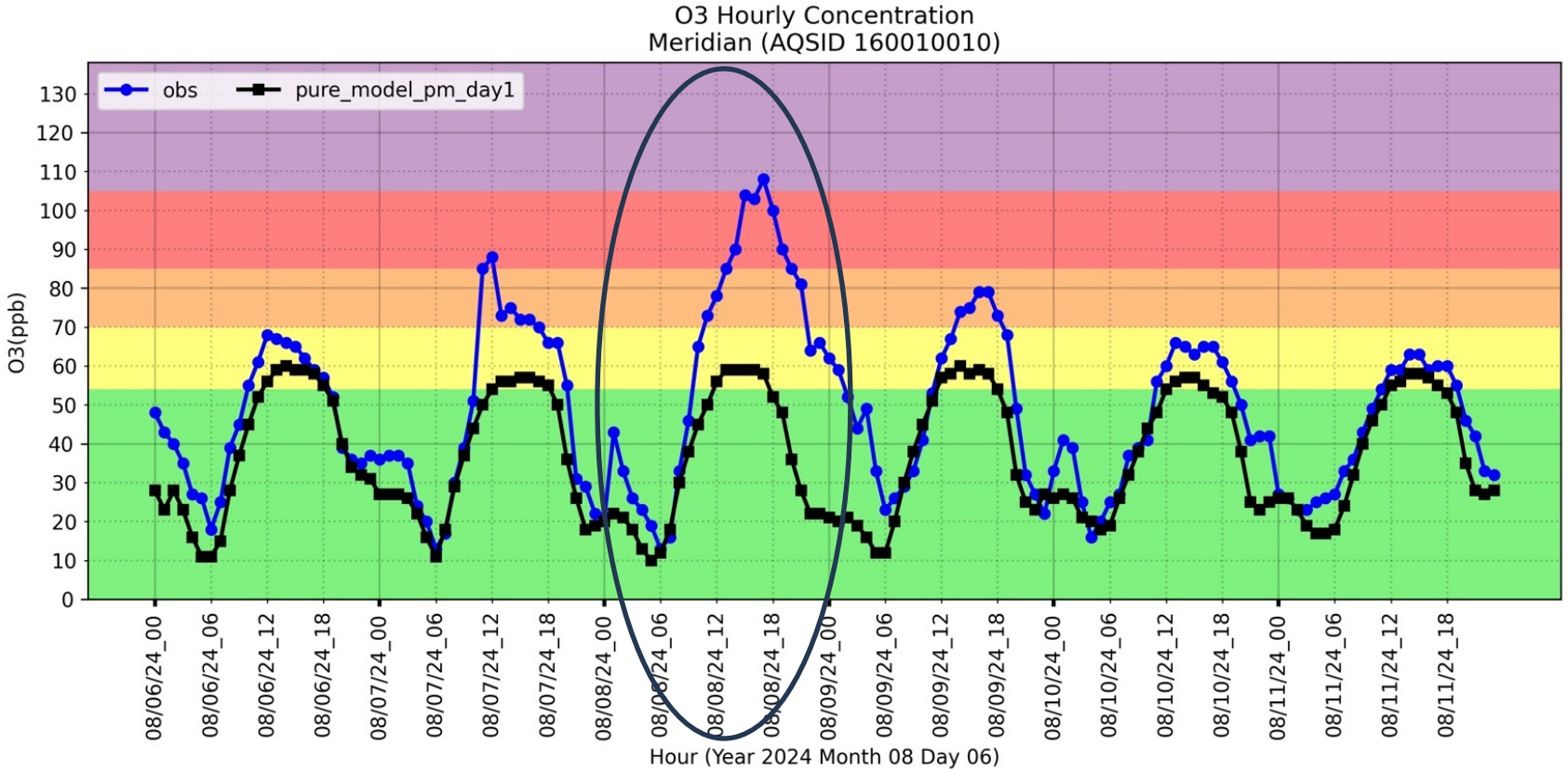
July 23 at Meridian Site (3) Wildfire Impact



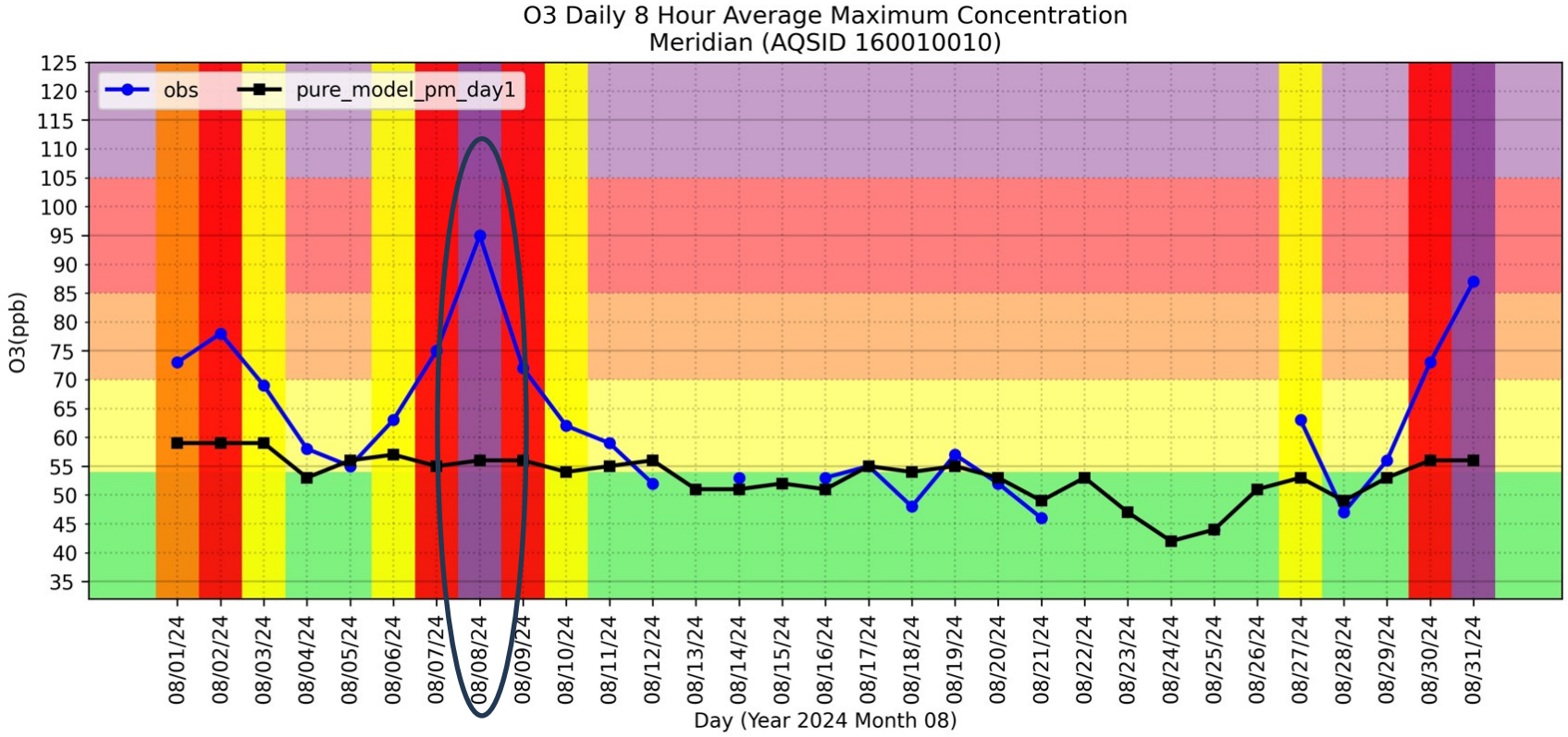
August 8 at Meridian Site (0) Wildfire Impact



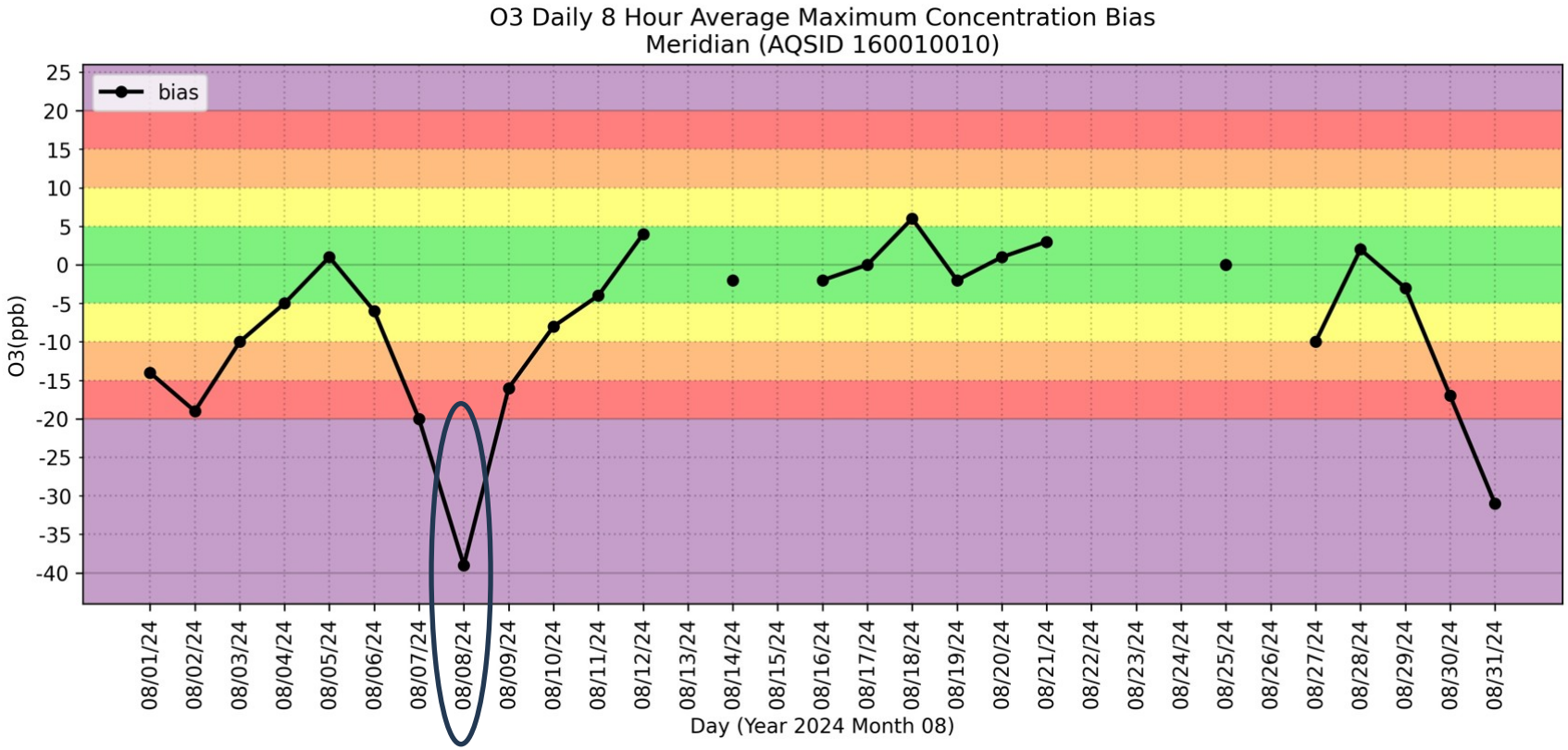
August 8 at Meridian Site (1) Wildfire Impact



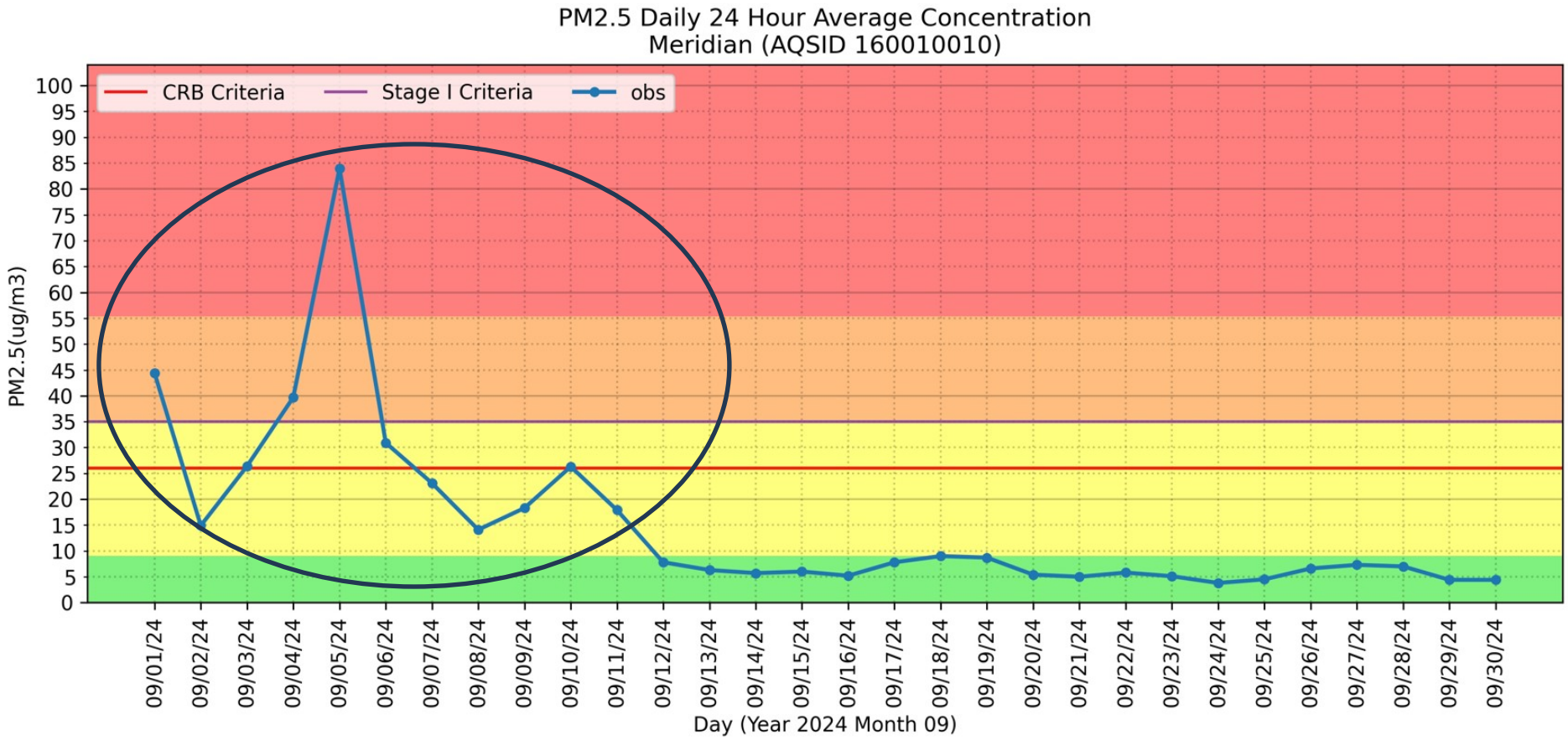
August 8 at Meridian Site (2) Wildfire Impact



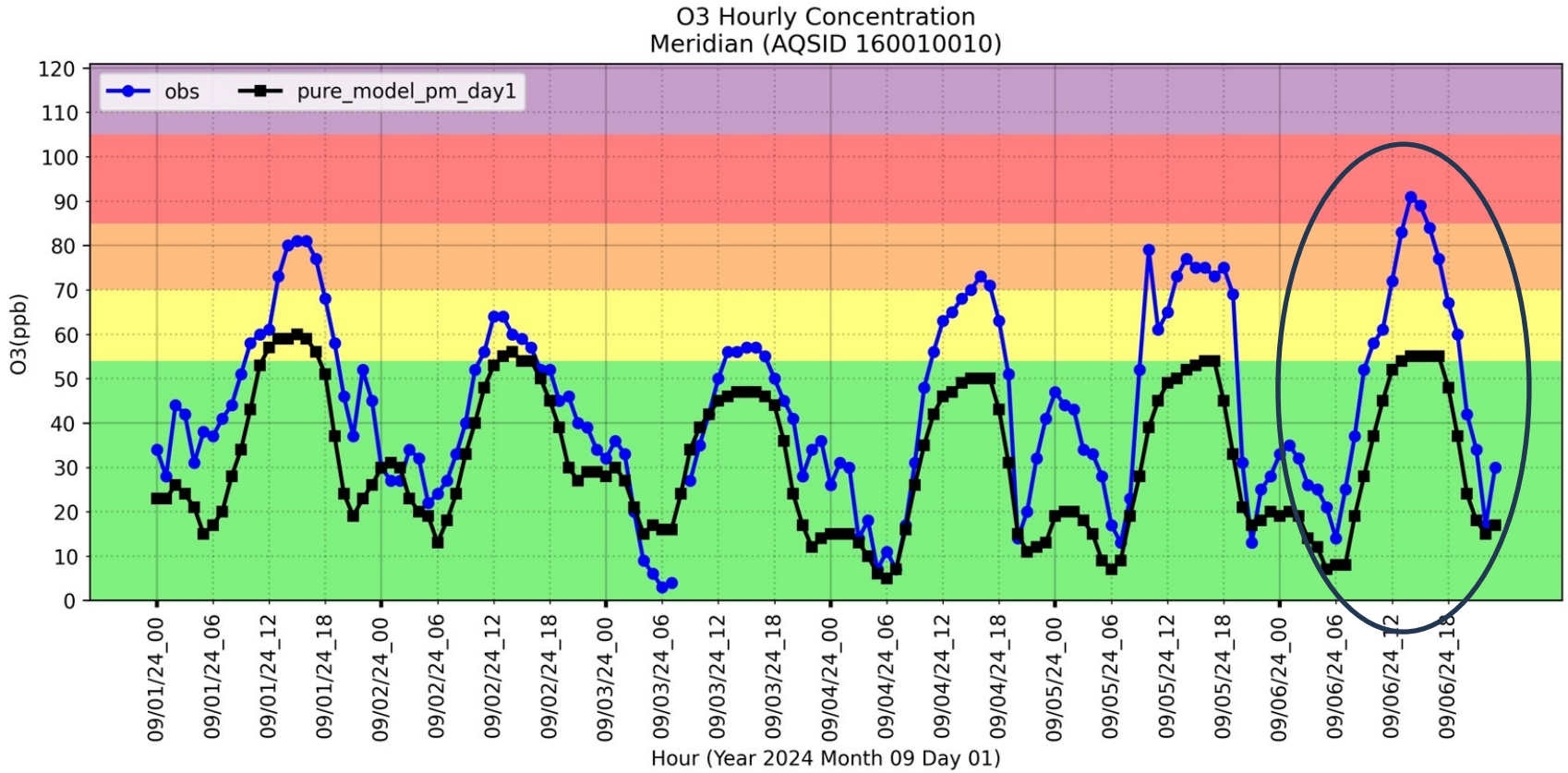
August 8 at Meridian Site (3) Wildfire Impact



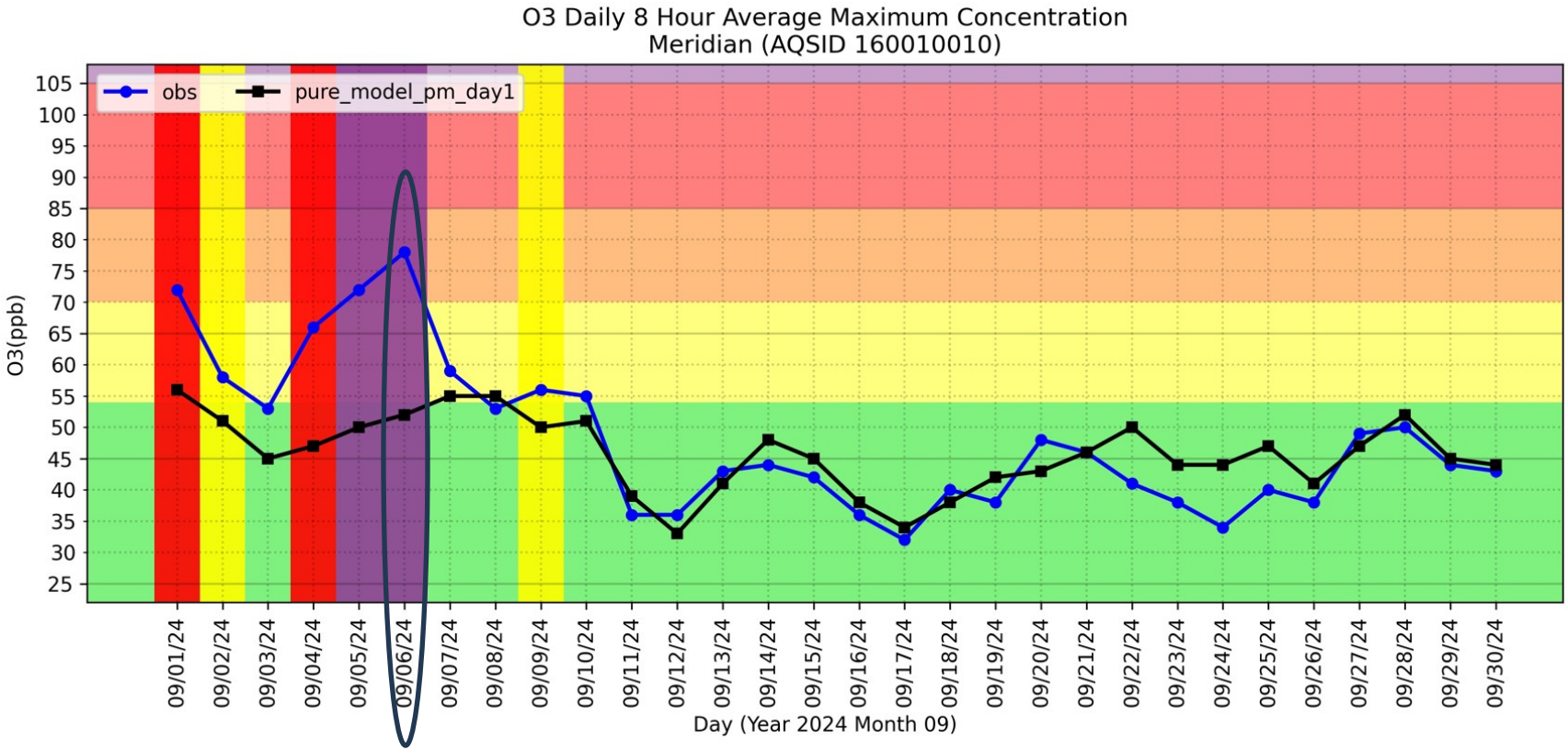
Sept. 6 at Meridian Site (0) Wildfire Impact



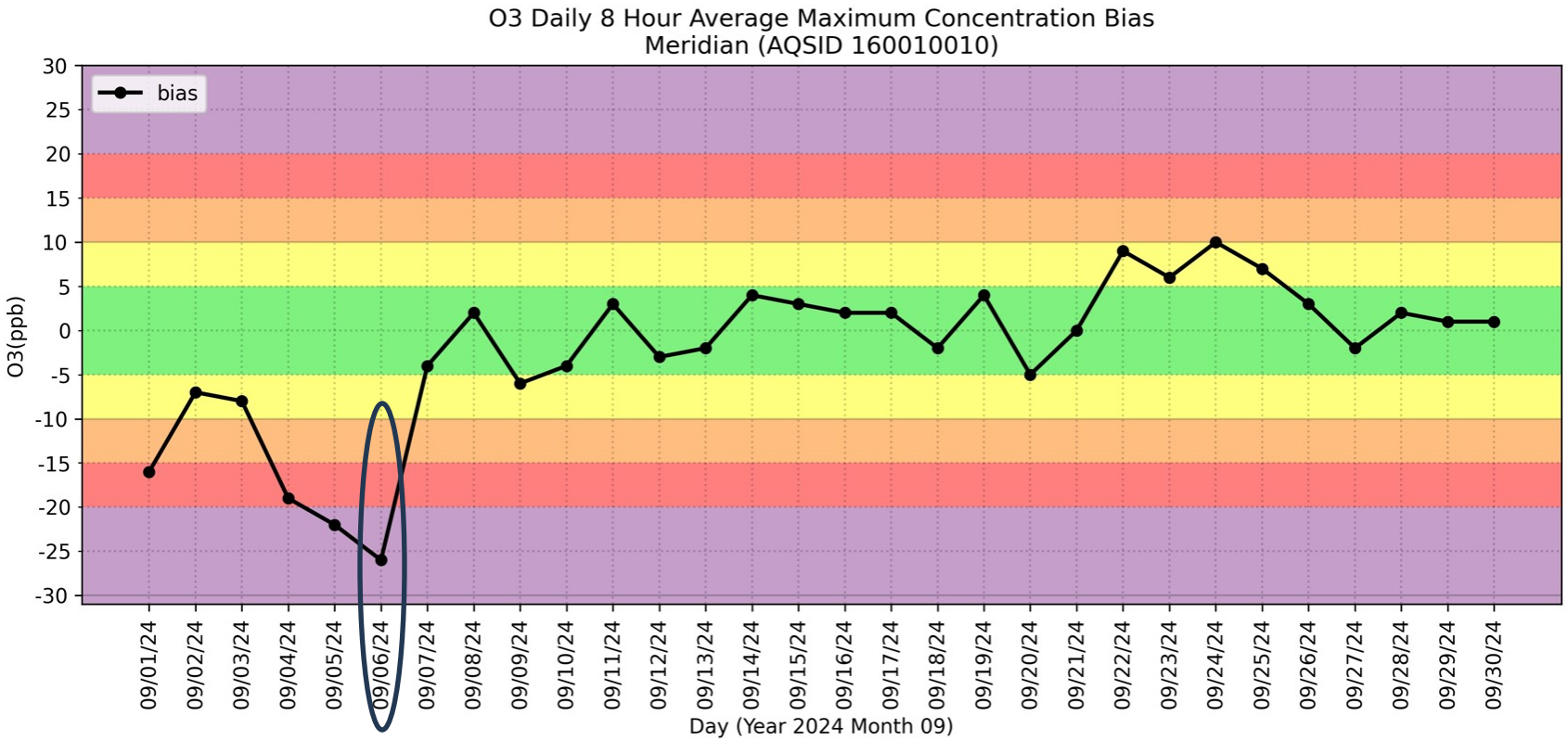
Sept. 6 at Meridian Site (1) Wildfire Impact



Sept. 6 at Meridian Site (2)



Sept. 6 at Meridian Site (3) Wildfire Impact



Summary

- Machine Learning Model is a good tool to screen/flag O3 Exceptional Events.

- Can be used as an evidence of Exceptional Events demonstration.

Questions and Discussion

The End